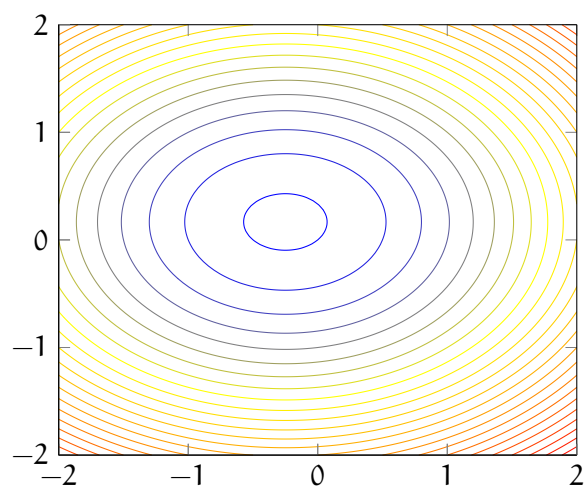
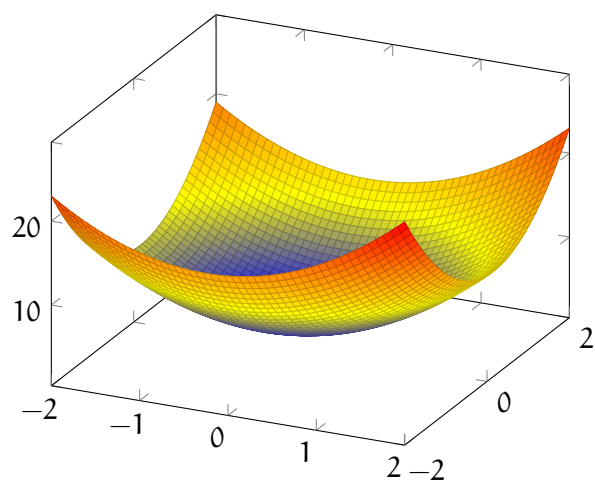


Consider the function

$$f(x_1, x_2) = 2x_1^2 + 3x_2^2 + x_1 - x_2 + 3$$

illustrated in the figures below.



Show that

$$x^* = \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{6} \end{pmatrix}$$

is a global minimum of f . Is it the only one?